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1. Project Title:

Maze Solving using General Search Tree (state space tree)

2. Introduction:

This project aims to implement an efficient algorithm for maze solving using the General search tree (State space tree) data structure. The approach will utilize the tree to efficiently track connected components within the maze and determine the optimal path from the start to the exit.

3. Problem Statement:

Given a 2D grid representing a maze with open and blocked cells, the objective is to find a path from the starting position to the exit efficiently.

4. Objectives:

- Develop an algorithm to efficiently determine whether a path exists between the start and exit.
- Visualize the maze and solution path using a graphical or text-based interface.
- Compare the efficiency of the General search tree (State space tree) approach with traditional search-based methods.

5. Methodology:

1. Maze Representation:

- Represent the maze as a **grid of cells**.

2. Maze Generation:

- Start with all walls intact and progressively remove them while ensuring connectivity.

3. Pathfinding using search tree:

- Initialize a General search tree (State space tree) where each cell starts as its own component.
- As walls are removed, merge adjacent cells into the same component using **trees**

6. Data Structures Used:

- **General search tree (State space tree):** For efficiently managing connected components in the maze.
- **Adjacency List / Adjacency Matrix:** To represent the connectivity of maze cells.

7. Expected Outcomes:

- A fully functional **maze solver** using the General search tree (State space tree)
- A **graphical or text-based representation** of the generated maze and its solution.

8. Tools & Technologies:

- **Programming Language:** Python.
- **Data Structures:** General search tree , Graphs

9. Timeline:

Task	Duration
Research and Literature Review	1 week
Maze Generator Implementation	1 weeks
General search tree Based Pathfinding	2 weeks
Testing and Debugging	1 week
Documentation & Final Report	1 week

10. Conclusion:

This project will demonstrate the effectiveness of the General search tree (State space tree) algorithm in solving mazes. By integrating maze generation, solving, and visualization, it will provide a complete understanding of how disjoint sets can be used for connectivity problems. The insights gained from performance comparisons will highlight the advantages and limitations of this approach compared to conventional pathfinding algorithms.